

Experiment 12:Tic Tac Toe game

EXPERIMENT: TIC TAC TOE GAME

Aim

To write a Python program to implement a Tic Tac Toe game using Artificial Intelligence techniques.

Objective

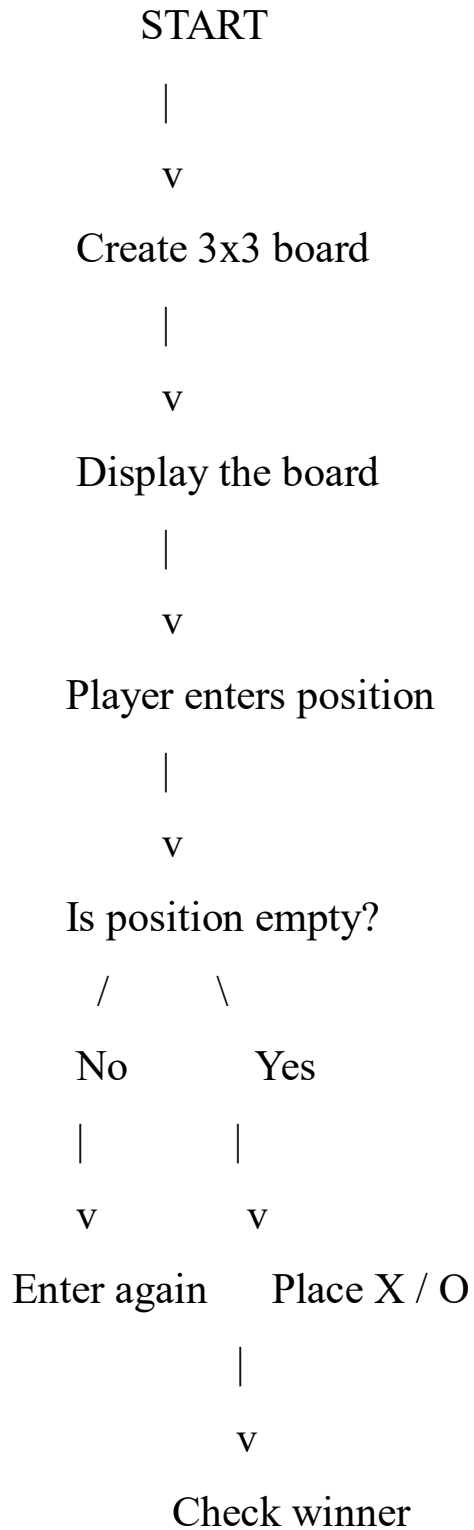
- To understand the basic concepts of game playing in Artificial Intelligence.
- To implement a two-player Tic Tac Toe game.
- To represent the game board using a Python list.
- To check winning, losing, and draw conditions.
- To display the final result of the game.

Algorithm

1. Start.
2. Create a 3×3 Tic Tac Toe board.
3. Initialize the board with empty spaces.
4. Display the board.
5. Ask the current player to enter a position.
6. Check whether the selected position is empty.
7. Place the player's symbol (X or O) on the board.
8. Check all rows, columns, and diagonals for a winning combination.
9. If a player wins, display the winner and stop.
10. If all positions are filled and there is no winner, declare a draw.

11. Otherwise, switch to the other player.
12. Repeat the process until the game ends.
13. Stop.

Flowchart:



```

        /      \
    Yes      No
    |        |
    v        v
Display winner Board full?

        /      \
    Yes      No
    |        |
    v        v
Display Switch
Draw player
    |        |
    v        |
STOP <----

```

Code (Python):

```
# Tic Tac Toe Game
```

```
board = [' ' for i in range(9)]
```

```
def display_board():
```

```
    print()
```

```
    print(board[0], '|', board[1], '|', board[2])
```

```
    print('--+---+--')
```

```
    print(board[3], '|', board[4], '|', board[5])
```

```
    print('--+---+--')
```

```
print(board[6], '|', board[7], '|', board[8])  
print()
```

```
def check_winner(player):
```

```
    winning_positions = [  
        (0, 1, 2),  
        (3, 4, 5),  
        (6, 7, 8),  
        (0, 3, 6),  
        (1, 4, 7),  
        (2, 5, 8),  
        (0, 4, 8),  
        (2, 4, 6)  
    ]
```

```
    for a, b, c in winning_positions:
```

```
        if board[a] == board[b] == board[c] == player:
```

```
            return True
```

```
    return False
```

```
def play_game():
```

```
    player = 'X'
```

```
    for turn in range(9):
```

```
display_board()
```

```
position = int(input(
    "Player " + player + ", enter position (1-9): "
)) - 1
```

```
if position < 0 or position > 8:
    print("Invalid position!")
    continue
```

```
if board[position] != ' ':
    print("Position already occupied!")
    continue
```

```
board[position] = player
```

```
if check_winner(player):
    display_board()
    print("Player", player, "wins!")
    return
```

```
if ' ' not in board:
    display_board()
    print("Game Draw!")
    return
```

```
if player == 'X':  
    player = 'O'  
else:  
    player = 'X'
```

play_game()

Output

```
| |  
--+---+--  
  
| |  
--+---+--  
  
| |
```

Player X, enter position (1-9): 1

```
X | |  
--+---+--  
  
| |  
--+---+--  
  
| |
```

Player O, enter position (1-9): 5

```
X | |
```

```

--+---+--
| O |
--+---+--

| |

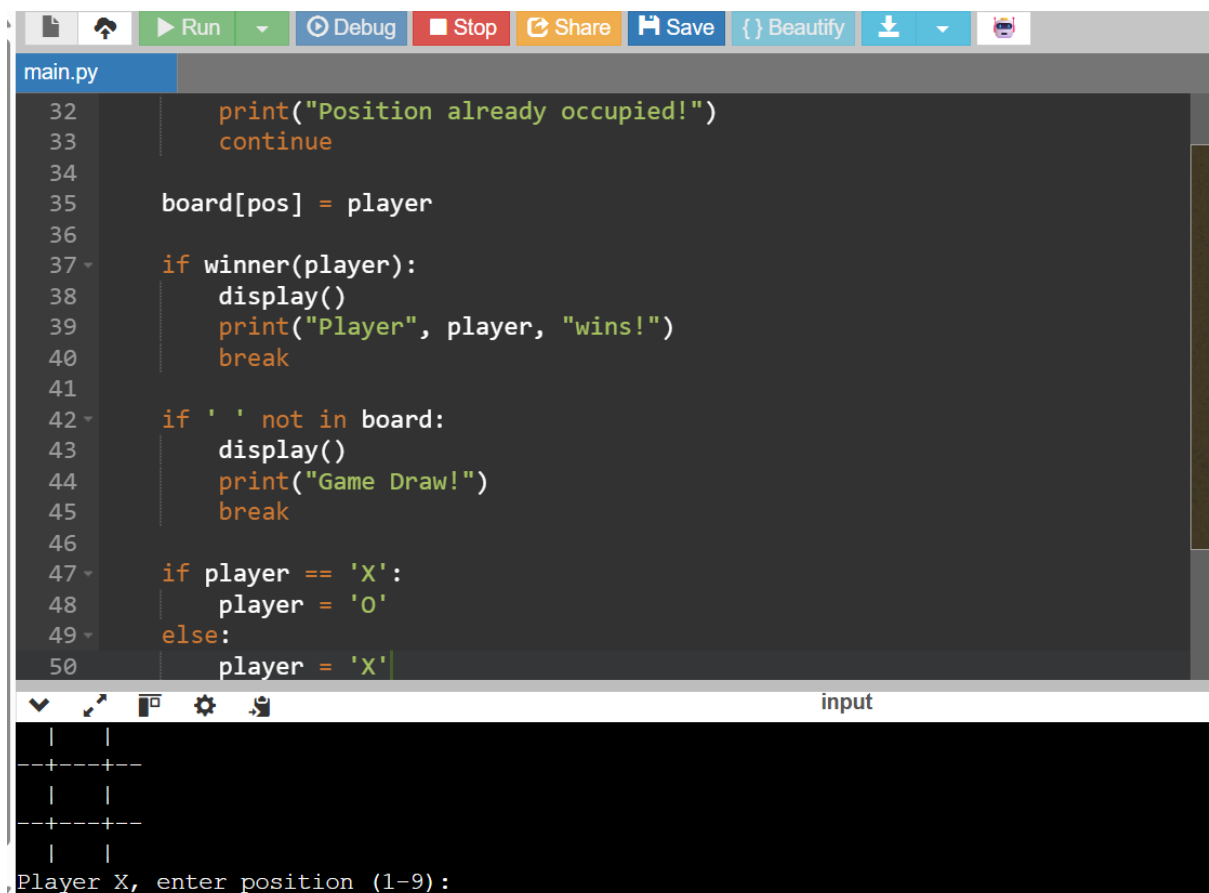
```

Player X, enter position (1-9): 2

...

Player X wins!

Output:



```

main.py
32     print("Position already occupied!")
33     continue
34
35     board[pos] = player
36
37     if winner(player):
38         display()
39         print("Player", player, "wins!")
40         break
41
42     if ' ' not in board:
43         display()
44         print("Game Draw!")
45         break
46
47     if player == 'X':
48         player = 'O'
49     else:
50         player = 'X'

```

input

```

| |
--+---+--
| |
--+---+--
| |
Player X, enter position (1-9):

```

Result:

Thus, the Python program to implement the Tic Tac Toe game was executed successfully, and the game correctly identified the winner or draw condition.

Conclusion:

The experiment helped in understanding the basic concepts of game playing in Artificial Intelligence. The Tic Tac Toe game was successfully implemented using Python by representing the board, accepting player moves, checking winning conditions, and determining the final result.